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Wollo University

Informatics College

Department of Information Technology

Integrated Project (IP, ADB and SAD)

Submission date December 25, 2018 E.C

Course Level Project for 3rd Year IT Students

The main objective of this project is to integrate courses (**Internet Programming**, **Database**, and **Software Engineering/SAD)** into a single, practical system.

By the end of the project, students should be able to:

1. Design and implement a **dynamic website using PHP**.
2. Implement a **relational database using MySQL**.
3. Prepare Software Development full documentation

Instruction:

- ☞ Prepare full SRS document as per the guideline given
- ☞ Identify main entities and relationships carefully, and Normalize database to avoid redundancy.
- ☞ Implement the database schema (Create tables with appropriate data types and constraints, implement relationships using key attributes, Perform CRUD operations through PHP scripts, implement transaction management where necessary, implement database backup and restore process using MySQL tools or PHP scripts)
- ☞ Develop the system based on functional requirements you have Analyzed.

Title:

Group1- Inventory and Billing Management System

Group 2: Online Student Result Management System

Group 3: Online Library Management System

Group 4: Online Voting System

Group 5: Online Job Portal

System Analysis and Design

Full System Development Documentation Guideline

1. General Information

- ✓ Wollo University Logo
- ✓ College
- ✓ Department
- ✓ Title Page
- ✓ Group Name
- ✓ Submission Date
- ✓ Approval Page (optional)
- ✓ Definitions, Acronyms, and Abbreviations
- ✓ Table of Contents
- ✓ Abstract

2. *Introduction*

- ✓ Background of the system
- ✓ Purpose of the System
- ✓ Scope

- ✓ Objectives
    - General objective
    - Specific Objectives
- ✓ System problems
- ✓ Limitation of the system
- ✓ Beneficiary of the system
- ✓ System Methodology
    - Data gathering Methodology
    - System Development Methodology (Development model used (Agile, Waterfall, Spiral, etc.)
- ✓ Feasibly Study
    - Economical
    - Political
    - Technical
    - Cost
    - Operational
- ✓ Significance of the system
- ✓ Budget of the system
- ✓ Schedule of the system

3. Software Requirements Specification (SRS)

- ✓ Proposed Solution
- ✓ System Workflow
- ✓ System Advantages
- ✓ System Business Rule
- ✓ Functional Requirements
- ✓ Non-Functional Requirements
- ✓ System Constraints

4. System Analysis

- ✓ Use-Case Diagram
- ✓ Use-Case Descriptions

- ✓ Context Diagram (DFD Level 0)
- ✓ DFD Level 1 & Level 2
- ✓ Entity-Relationship Diagram (ERD)
- ✓ Activity Diagrams
- ✓ Sequence Diagrams System Design
- ✓ System Architecture
- ✓ Class Diagram
- ✓ Database Design
- ✓ User Interface (UI) Design
- ✓ Data Dictionary

5. Implementation Documentation

- ✓ Tools Used
- ✓ System Modules
- ✓ Coding Standards
- ✓ Important Code Snippets

6. Testing Documentation

- ✓ Test Plan
- ✓ Test Cases
- ✓ Unit Testing
- ✓ Integration Testing
- ✓ System Testing
- ✓ User Acceptance Testing (UAT)

7. Deployment Documentation

- ✓ Installation Guide
- ✓ Deployment Diagram
- ✓ Backup and Recovery Plan

8. User Manual

- ✓ How to use the system with screenshots

9. Maintenance Documentation

- ✓ System update and bug-fix procedures

10. Conclusion & Future Work

- ✓ Summary, limitations, recommendations

11. Appendices

- ✓ Additional diagrams, sample data, forms, code appendix